

QUALITY ASSURANCE · TEST PLAN

Booking.com Car Rental

New York Pick-up Location — Web Application

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Application:	booking.com/cars/index.html
Scope:	Car Rental Search & Booking (NYC)
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1. Project / Test Sub-process

Project: Booking.com — Car Rental Vertical

Test Sub-process: System & Functional Testing of the Car Rental search, filtering, and booking flow for the *New York, NY (United States)* pick-up location, accessed via <https://www.booking.com/cars/index.html>.

The car rental section allows users to search for rental vehicles by location, pick-up & drop-off date and time, filter results by price, supplier, car category and policies, and complete a booking. This plan covers the end-to-end user journey for the New York market.

2. Test Item(s)

- Search form (location, dates, times, "Driver aged 30 – 65")
- Location autocomplete & geolocation suggestions for New York
- Pick-up and drop-off date/time pickers
- Search Results Listing Page (SRP)
- Sort controls (price, recommended, supplier rating)
- Filter panel (price range, car type, supplier, fuel policy, transmission, mileage)
- Vehicle details page
- Rental conditions & what's included
- Extras selection (GPS, child seats, additional driver)
- Booking summary & price breakdown (incl. taxes, fees)
- Driver details form & validation
- Payment page integration
- Booking confirmation & email notification
- Currency switcher (USD / EUR / UAH / etc.)
- Language switcher (EN / DE / ES / UA / etc.)
- Responsive layout (desktop, tablet, mobile)
- "Manage your bookings" entry point

3. Test Scope

In Scope

- **Functional testing** of the full booking flow for pick-up in New York (JFK, LGA, EWR, and downtown Manhattan locations).
- **UI / UX testing** on Chrome, Firefox, Safari, and Edge (latest two versions).
- **Cross-device testing** on desktop (1920×1080, 1440×900), tablet (iPad), and mobile (iPhone 14, Pixel 7).
- **Localization smoke tests** for English, with spot-checks for Ukrainian and Spanish translations.
- **Validation testing** for date/time, age restrictions, mandatory fields.
- **Negative testing** (invalid dates, past dates, unsupported age ranges).
- **Compatibility & accessibility smoke (WCAG 2.1 AA — keyboard navigation, alt text, contrast).**
- **Basic API checks** for the search and pricing endpoints (status codes, response schema).

Out of Scope

- Performance / load testing of Booking.com backend infrastructure.
- Penetration testing & deep security audits (handled by Booking.com Security team).
- Real money transactions — payment is verified using sandbox / test cards only.
- Native iOS and Android Booking.com applications.
- Pick-up locations outside the New York metropolitan area.
- Third-party rental supplier internal systems (Hertz, Avis, Budget back-office).

4. Assumptions and Constraints

Assumptions

- The staging/UAT environment of Booking.com Cars will be available throughout the test window.
- Test supplier inventory for New York is seeded and consistent across test runs.
- A sandbox payment gateway (e.g. Adyen test mode) is available; test card numbers will be provided.
- Stable internet connection is available on all test devices.
- Test accounts with various states (new user, returning user, Genius member) are pre-provisioned.

Constraints

- Limited access to production data — real customer data must not be used.
- Third-party supplier APIs have rate limits; large parallel runs are restricted.
- Project timeline: 4 weeks of active testing (see §23 Schedule).
- Test team capacity: 1 QA Lead + 2 QA Engineers (see §20).

5. Stakeholders

Role	Name	Responsibility
Product Owner	TBD	Approves scope, accepts deliverables
Project Manager	TBD	Timeline, coordination, reporting
QA Lead / Test Manager	Dmytro Radchenko	Owens this Test Plan, leads execution
QA Engineers	TBD x2	Design and execute test cases
Development Lead	TBD	Defect resolution, technical questions
UX / UI Designer	TBD	Design specs, UI clarifications
Business Analyst	TBD	Requirements clarification
DevOps	TBD	Environment availability
End Users (UAT)	TBD	User acceptance feedback

6. Testing Communication

Activity	Channel	Cadence	Audience
Daily stand-up	MS Teams / Zoom	Daily, 15 min	Dev + QA team
Test status report	Email + Confluence	Weekly (Fridays)	All stakeholders
Defect triage	Jira + Slack	Twice a week	QA Lead, Dev Lead, PM
Sprint demo	Zoom	End of sprint	Product Owner, Team
Ad-hoc questions	Slack channel #cars-ny-qa	Continuous	Team
Test closure report	Confluence + Email	End of project	All stakeholders

7. Product Risks

ID	Risk	Likelihood	Impact	Mitigation
PR-01	Search returns no results for valid NY pick-up location	Medium	High	Test multiple NY airports + downtown, monitor supplier feed
PR-02	Total price shown $\neq$ price charged at payment step	Medium	Critical	End-to-end price verification on every test run
PR-03	Date picker allows past or invalid date ranges	Low	High	Boundary value & negative tests on date inputs
PR-04	Localization breaks layout (long German strings, RTL languages)	Medium	Medium	Visual regression for top 5 languages
PR-05	Booking confirmation email not received	Low	High	Verify SMTP integration; check spam folders
PR-06	Currency conversion rounding errors	Medium	Medium	Cross-check displayed totals with exchange-rate API
PR-07	Mobile layout overlaps or hides "Book" CTA	Medium	High	Responsive tests on 3+ viewports
PR-08	Accessibility issues (low contrast, missing labels)	Medium	Medium	Run axe-core scans, manual screen-reader checks

8. Project Risks

ID	Risk	Likelihood	Impact	Mitigation
PJ-01	Staging environment instability	Medium	High	Daily health-check; rollback procedure agreed with DevOps
PJ-02	Late delivery of requirements / UI mockups	Medium	Medium	Risk-based test design; iterative case refinement
PJ-03	Insufficient QA capacity	Low	High	Cross-training; explore short-term contractors
PJ-04	Third-party supplier feed unavailable	Medium	Medium	Stub/mockbed supplier responses for offline runs
PJ-05	Scope creep (additional cities requested)	Medium	Medium	Strict change-control via PM

9. Test Strategy for Test Sub-process

The test approach follows a **risk-based, shift-left strategy** and combines manual and automated testing across the test pyramid.

Test Levels

- **Unit tests** — owned by development; QA reviews coverage reports.
- **Integration tests** — API contract tests on search, pricing, and booking endpoints (Postman / RestAssured).
- **System tests** — end-to-end booking journey on staging.
- **User Acceptance Tests (UAT)** — selected scenarios run with the Product Owner.

Test Types

- Functional, UI, Compatibility, Responsive, Localization, Accessibility (WCAG 2.1 AA smoke), Regression, Smoke, Sanity.

Automation

- Smoke + regression suite automated in **Playwright (TypeScript)**.
- API checks automated in **Postman + Newman** integrated into CI.
- Target $\geq 60\%$ automation coverage of regression scope by project end.

10. Test Deliverables

- This Test Plan document (v1.0+).
- Test cases & checklists in TestRail (or equivalent).
- Automated test scripts (Playwright) committed to the QA repository.
- Test data sets and supplier mock responses.
- Defect reports in Jira.
- Daily and weekly test execution reports.
- Traceability matrix (requirements ↔ test cases ↔ defects).
- Final Test Summary / Closure Report.

11. Test Design Techniques

Technique	Where Applied
Equivalence Partitioning	Driver age, rental duration, price filter ranges
Boundary Value Analysis	Date picker (today, today-1, max +1 year), age (18, 25, 30, 65, 75)
Decision Table Testing	Booking eligibility (age × license × payment method × supplier policy)
State Transition Testing	Booking states: searched → selected → reserved → paid → confirmed → cancelled
Use Case / Scenario Testing	Full traveler journey for NY pick-up
Exploratory Testing	Time-boxed charters around filters, sort, and edge cases
Error Guessing	Common UX pitfalls — empty cart, browser back, double-click on "Book"
Checklist-based Testing	Localization & accessibility passes

12. Test Completion (Exit) Criteria

- 100% of planned high-priority test cases executed.
- ≥ 95% pass rate on high-priority cases; 100% on critical-path cases.
- 0 open Critical or High defects; ≤ 5 open Medium defects with approved workarounds.
- All Severity-1 defects retested and closed.
- Regression suite executed successfully on the final release candidate.
- Test Summary Report signed off by QA Lead and Product Owner.

13. Metrics Collected

- Test case execution progress (planned / executed / passed / failed / blocked).
- Defect density per module.
- Defect severity & priority distribution.
- Defect status aging (open vs. closed over time).
- Defect leakage (defects found in UAT / Production).
- Test automation coverage %.
- Automated test pass rate & flakiness rate.
- Requirements coverage (via traceability matrix).
- Effort variance (planned vs. actual hours).
- Environment downtime hours.

14. Test Data Requirements

- **User accounts:** guest user, returning user, Genius Level 1/2/3 user.
- **Drivers:** ages 18, 24, 25, 30, 65, 70 — to cover age-based pricing rules.
- **Locations:** JFK Airport, LaGuardia (LGA), Newark (EWR), Manhattan downtown, Brooklyn.
- **Date ranges:** 1-day, 3-day, 7-day, 30-day rentals; weekends, holidays, peak season.
- **Payment:** sandbox Visa, Mastercard, Amex (success + decline scenarios); 3-D Secure flow.
- **Currencies:** USD, EUR, GBP, UAH.
- **Languages:** EN, ES, DE, UA.
- **Supplier mix:** at least 3 different suppliers (e.g., Hertz, Enterprise, Sixt) to validate variability.
- **PII handling:** all data is synthetic; no production PII is used.

15. Test Environment Requirements

Environments

- **DEV** — for smoke after deploys, unstable allowed.
- **QA / Staging** — primary test environment, mirror of production.
- **UAT** — used for stakeholder acceptance.

Browsers & Devices

- Google Chrome (latest, latest-1)
- Mozilla Firefox (latest)
- Apple Safari (latest, macOS & iOS)
- Microsoft Edge (latest)
- Desktop: 1920×1080, 1440×900
- Tablet: iPad (10.9") portrait & landscape
- Mobile: iPhone 14 (iOS 17+), Pixel 7 (Android 14)

Tooling

- Test management: **TestRail** (or Xray for Jira)
- Defect tracking: **Jira**
- Automation: **Playwright** (TypeScript), **Postman** / **Newman**
- CI/CD: **GitHub Actions**
- Accessibility: **axe DevTools**, screen readers (VoiceOver, NVDA)
- Visual regression: **Percy** or **Playwright snapshots**
- Network throttling: Chrome DevTools (Slow 3G, Fast 3G profiles)

16. Retesting and Regression Testing

Retesting (Confirmation Testing)

Every defect marked "Ready for retest" is re-executed using the original failing steps. A defect is closed only after it has been verified as fixed in the same environment where it was found.

Regression Testing

- Smoke regression (≈ 30 cases) — runs automatically on every staging deploy.
- Full regression suite (≈ 250 cases) — runs nightly and before each release.
- Targeted regression — selected by impact analysis around changed modules.
- Visual regression — Percy snapshots compared on every PR affecting UI.

17. Suspension and Resumption Criteria

Suspension Criteria — testing is paused if

- The staging environment is unavailable for > 4 working hours.
- A blocker defect prevents the main booking flow from being executed end-to-end.
- Critical test data (supplier feed, payment sandbox) is unavailable.
- A new release breaks more than 30% of the smoke suite.

Resumption Criteria — testing resumes when

- Environment is verified stable via a green smoke run.
- Blocker defects are fixed, retested, and closed.
- Test data integrity is restored.
- QA Lead formally announces resumption in the project channel.

18. Deviations from the Organizational Test Strategy

- Performance and security testing are **delegated to dedicated teams** and are not part of this sub-process — a deviation from the standard "full-stack quality" approach.
- Native mobile applications (iOS / Android) are out of scope for this iteration.
- Automation coverage target is set at 60% (organizational baseline is 70%) due to the limited project window.

19. Testing Activities and Estimates

Activity	Effort (person-days)
Test planning & documentation	3
Test case design	6
Test data preparation	2
Environment & tooling setup	2
Manual test execution (2 cycles)	10
Automation scripting (Playwright + API)	8
Regression execution	3
Defect retesting & verification	3
UAT support	2
Reporting & closure	2
Total	≈ 41 person-days

20. Roles, Activities, and Responsibilities

Role	Key Activities	Responsibilities
QA Lead / Test Manager	Planning, reporting, defect triage	Overall QA delivery, sign-off
Senior QA Engineer	Automation, complex scenarios	Playwright framework, CI pipeline
QA Engineer	Manual execution, exploratory testing	Test case execution, defect logging
Developer	Fix defects, unit tests	Code quality, build stability
Product Owner	UAT, acceptance	Business sign-off
DevOps	Environment, CI/CD	Uptime, deployments

21. Hiring Needs

No new permanent hires are planned. A **short-term contractor** (1 automation engineer, ~3 weeks) may be engaged if the automation backlog exceeds capacity by > 20%. Decision point: end of Week 2.

22. Training Needs

- **Playwright** refresher (½ day) — for QA engineers new to the framework.
- **Accessibility testing** (½ day) — WCAG 2.1 AA basics + axe DevTools.
- **Booking.com domain knowledge** (1 day) — walkthrough with the BA covering car-rental business rules.
- Optional: **API testing with Postman** (½ day).

23. Schedule

Phase	Activities	Window
Week 1	Test planning, environment setup, test data preparation, case design (high-priority)	May 18 – May 22, 2026
Week 2	Manual test execution — Cycle 1, automation framework setup, defect logging	May 25 – May 29, 2026
Week 3	Manual test execution — Cycle 2, regression scripting, defect retesting	Jun 1 – Jun 5, 2026
Week 4	Final regression, UAT support, closure report, sign-off	Jun 8 – Jun 12, 2026

Revision Loop: Confirm if further adjustments are needed. This plan is a living document and will be updated at the end of each sprint based on findings, defect trends, and stakeholder feedback.

— End of Test Plan —